

Experiment 8

Execute and display the results for the following Nmap service version and OS detection commands in Kali Linux Operating System:

- i) Detect the version of services running
- ii) Aggressive scan
- iii) Detect operating system of the target

Aim

To perform service version detection, aggressive scanning, and operating system detection using Nmap in Kali Linux.

Procedure

1. Open the Kali Linux Terminal.
2. Select an authorized target system or scan your own machine.
3. Use the Nmap service version detection option to identify running service versions.
4. Perform an aggressive scan to collect detailed information about the target.
5. Use Nmap OS detection to identify the target operating system.
6. Observe and record the scan results.

```
$ pwsh
PowerShell 7.6.2

(kali@kali)-[/home/kali]
PS> nmap -sV 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 03:52 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000050s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds

(kali@kali)-[/home/kali]
PS> nmap -A 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-03 03:52 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.000071s latency).
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)
Too many fingerprints match this host to give specific OS details
Network Distance: 0 hops

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.74 seconds

(kali@kali)-[/home/kali]
PS> sudo nmap -O 127.0.0.1
[sudo] password for kali:
```

Result

Thus, Nmap service version detection, aggressive scanning, and operating system detection were successfully performed in Kali Linux.

